

MATERIAL SAFETY DATA SHEET

1. Chemical Product Identification

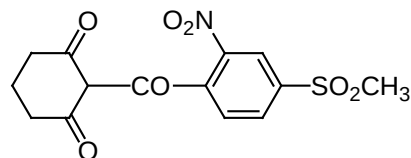
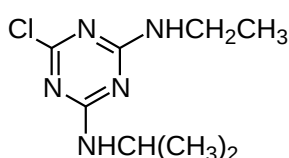
Product Name: MESOTRIONE + ATRAZINE 8%+80.8% WG

Active ingredient: Atrazine 80.8% Mesotrione 8%

Molecular Formula: $C_8H_{14}ClN_5$ $C_{14}H_{13}NO_7S$

Molecular Weight: 215.7g/mol 339.3g/mol

Structural Formula:



IUPAC 6-chloro-*N*-ethyl-*N'*-(1-methylethyl)-1,3,5-triazine-2,4-diamine 2-(4-mesyl-2-nitrobenzoyl)cyclohexane-1,3-dione

Chemical Group: Triazines herbicides Benzoylcyclohexanedione
Herbicide

CAS No.: 1912-24-9 104206-82-8

Manufacturer: Shandong Binnong Technology Co., Ltd.

Address: No.518, Yongxin Road, Binbei Town, Binzhou, Shandong, China

2. Hazards Identification

Symptoms of acute exposure:

Exposure may cause irritation of the eyes, skin, or respiratory tract. If swallowed, nausea, vomiting, or diarrhea can result. A skin sensitization (allergic) reaction may occur in sensitive individuals.

Carcinogenicity: NTP/!ARC/OSHA Carcinogen listing 10/94 No

Effects of overexposure: Dermatitis, conjunctivitis, muscular weakness, hyperactivity, ptosis, dyspnea, convulsions.

3. Composition / Information on Ingredients

Ingredient		CAS No.	Content, % (w/v)
Active ingredient	Atrazine	1912-24-9	80.8%
Active ingredient	Mesotrione	104206-82-8	8%

4. First Aid Measures

If poisoning is suspected, immediately contact a physician, the nearest hospital, or nearest Poison Control Center. Tell the person contacted the complete product name, and the type and amount of exposure. Describe any symptoms and follow advice given.

First aid if SWALLOWED: If victim is fully conscious, immediately give 1 or 2 glasses of water to drink and induce vomiting. Never give anything by mouth to an unconscious person. Apply artificial respiration if necessary. Seek medical attention immediately.

First Aid for EYE Exposure: Immediately flush eyes with large amounts of running water. Hold eye lids apart to rinse the entire surface of the eye and lids. Do not apply any medicating agents except on the advice of a physician. Get medical attention immediately if irritation persists.

First Aid for SKIN Exposure: Remove contaminated clothing and wash before reuse. Wash thoroughly, with plenty of soap and water, including hair and under fingernails. Do not apply any medicating agents except on the advice of a physician. Get medical attention immediately if irritation persists.

First Aid for INHALATION: Remove from exposure. Seek medical attention. Apply artificial respiration if necessary.

Note to PHYSICIAN: There is no specific antidote if ingested. If a large amount has been ingested and emesis has been inadequate, lavage stomach. Five mg/kg of activated charcoal suspension can be given to absorb the remaining toxicant.

5. Fire-Fighting Measures

Flash point and method: > 200°F (TCC) (Non-Combustible)

Upper and lower flammable (explosive) limits in air: Not applicable.

Auto-ignition temperature: Not applicable.

Unusual Fire, Explosion and Reactivity Hazards

During a fire, irritating and possibly toxic gases may be generated by thermal decomposition or

combustion.

In Case of Fire

Use dry chemical, foam or CO₂ extinguishing media. Wear full protective clothing and self-contained breathing apparatus. Evacuate nonessential personnel from the area to prevent human exposure to fire, smoke, fumes or products of combustion. Prevent use of contaminated buildings, area, and equipment until decontaminated. Water runoff can cause environmental damage. If water is used to fight fire, dike and collect runoff.

6. Accidental Release Measures

In Case of Spill or Leak

Control the spill at its source. Contain the spill to prevent from spreading or contaminating soil or from entering sewage and drainage systems or any body of water. Clean up spills immediately, observing precautions outlined in Section 8. Sweep up material and place in a compatible disposal container. Scrub area with hard water detergent (e.g. commercial products such as Tide, Joy, Spic and Span). Pick up wash liquid with additional absorbent and place into compatible disposal container. Once all material is cleaned up and placed in a disposal container, seal container and arrange for disposition.

7. Handling and Storage

Store the material in a well-ventilated, secure area out of reach of children and domestic animals. Do not store food, beverages or tobacco products in the storage area. Prevent eating, drinking, tobacco use, and cosmetic application in areas where there is a potential for exposure to the material. Wash thoroughly with soap and water after handling.

8. Exposure Controls/Personal Protection

The following recommendations for exposure controls/personal protection are intended for the manufacture, formulation, packaging and use of this product.

For commercial applications and/or on-farm applications consult the product label.

Personal protective equipment for each exposure route:

General: Avoid breathing dust, vapours or aerosols. Avoid contact with eye, skin and clothing.

Wash thoroughly after handling and before eating, drinking, or handling tobacco.

INGESTION: Prevent eating, drinking, tobacco usage and cosmetic application in areas where there is a potential for exposure to the material. Wash thoroughly with soap and water after handling.

EYES: Where eye contact is likely, use chemical splash goggles. Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower.

SKIN: Where contact is likely, wear chemical-resistant (such as nitrile or butyl) gloves, coveralls, socks and chemical-resistant footwear. For overhead exposure, wear chemical-resistant headgear.

INHALATION: A respirator is not normally required when handling this substance. Use effective engineering controls to comply with occupational exposure limits.

In case of emergency spills, use a NIOSH approved respirator with any N, R, P or HE filter.

Handling Precautions: Handle carefully so as to avoid container damage. Keep material in original containers. Store away from foodstuffs.

Storing Precautions: Do not put leaking packages into storage. Store at temperatures not exceeding 105°F.

Other Precautions: Keep out of reach of children. Avoid contamination of feed and foodstuffs. Avoid contamination of water supplies, streams, and ponds

9. Physical and Chemical Properties

Appearance	White solid
Odor	odorless
Specific Gravity:	1.13 at 20°C(Typical)
Solubility In Water:	33 mg/l (pH 7, 22°C) (Atrazine Technical). 2.2 g/l In water 2.2 (pH 4.8) (20°C) (Mesotrione Technical)
Suspension rate (%):	≥ 90.0
Boiling Point:	103°C
Vapor Pressure:	1.5 mmHg 22°C
pH:	6-9

10. Stability and Reactivity

Stability:	Stable under normal use and storage conditions.
Hazardous Polymerization:	Will not occur.
Conditions to Avoid:	Excessive temperatures
Hazardous Decomposition Products:	Carbon monoxide, hydrogen cyanide, acetonitrile
Materials to Avoid:	Acids and alkalies.

11. Toxicological Information

Acute oral toxicity:	LD ₅₀ for rats >2 000 mg/kg body weight
Acute dermal Toxicity:	LD ₅₀ for rat > 2000 mg/kg body weight
Acute inhalation toxicity:	LC ₅₀ for rat>2.1 mg/l air 4-h
Acute eye irritation:	non- irritant to eye of rabbit
Acute dermal irritation:	Non- irritant to skin of rabbit
Skin sensitization:	weak sensitiser to skin fo guinea pig

12. Ecological Information

Birds:	Acute oral LC ₅₀ for bobwhite quail, Japanese quail, ring necked pheasant and mallard duck 5 000 mg/kg Acute oral LC ₅₀ for peking ducks 10 000 mg/kg
Fish:	Acute toxicity for rainbow trout, carp, bluegill, and guppies: the (96 h) LC ₅₀ varied between 4.3 mg/kg (guppies) and 100 mg/kg(carp).
Bees:	LD ₅₀ (oral) > 97 µg / bee.
Daphnia:	EC ₅₀ (48 hr) was 6.9 mg/kg. The EC ₅₀ (21 d) was 0.12 mg/kg
Algae:	Scenedesmus subspicatus(72 h): 0,043 mg/l Selenastrum capricornutum(72 h):0,13 mg/l
Earthworms:	Low toxicity for earthworms

13. Disposal Considerations

Waste disposal information: Do not reuse product containers. Dispose of product containers, waste containers, and residues according to local, state, and federal health and environmental

regulations.

Characteristic Waste: Under certain circumstances, discarded product may exhibit TCLP hazardous characteristics. A hazardous waste determination should be done on a case by case basis.

Listed Waste: Not Applicable

14. Transport Information

Shipping information such as shipping classification:

Transportation of dangerous goods classification - road/rail not regulated.

Hazard Class or Division: 9

Identification Number: 3077

Packing Group: III

15. Regulatory Information

WHMIS classification for product: Exempt

A statement that the MSDS has been prepared to meet WHMIS requirements, except for use of the 16 headings.

This MSDS has been prepared in accordance with WHMIS requirements, but the data are presented under 16 headings. Other regulations; restrictions and prohibitions.

16. Other Information

All information and instructions provided in this Material Safety Data Sheet (MSDS) are based on the current state of scientific and technical knowledge at the date indicated on the present MSDS and are presented in good faith and believed to be correct. This information applies to the PRODUCT AS SUCH. In case of new formulations or mixes, it is necessary to ascertain that a new danger will not appear.

It is the responsibility of persons in receipt of this MSDS to ensure that the information contained herein is properly read and understood by all people who may use, handle, dispose or in any way come in contact with the product. If the recipient subsequently produces formulations(s) containing this product, it is the recipients sole responsibility to ensure the transfer of all relevant

information from this MSDS to their own MSDS.

ORIGINAL CONTROLLED